

sercon

User Manual

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sercon manual

Long-form reference for the `pkg/scriptengine` library, the `sercon` CLI, the built-in script API, and the JavaScript runtime surface that `scriptengine` exposes via [goja](#) and [goja_nodejs](#). Examples are runnable — save snippets as `.ts` files and execute with `sercon file.ts` unless otherwise noted.

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1. Overview

`sercon` is a Go program that runs TypeScript files. It's built from two parts:

- `pkg/scriptengine` — a library you embed in your own Go code. You register Go-callable bindings, then call `Run` / `RunFile` to execute a `.ts` script that talks to them.
- `cmd/sercon` — a thin CLI built on the library, useful for ad-hoc test scripts and as a working example of every binding kind.

The runtime is pure Go: [goja](#) is the JS engine, [esbuild](#) (used as a Go library) is the TS→JS transpiler, and [goja_nodejs](#) provides Promises, `setTimeout`, `console`, and CommonJS `require`. No cgo, no Node.

2. Quickstart

Install:

```
go install github.com/codedeviate/sercon/cmd/sercon@latest
```

Run one of the bundled examples:

```
sercon examples/scripts/smoke.ts examples/scripts/async.ts
```

...or run them all via `make demo`. `examples/scripts/` ships a runnable `.ts` (or `.tsx`) file per feature area — `hash.ts`, `strings.ts`, `path-and-time.ts`, `default-export.ts`, `tsx-demo.ts`, and the original `smoke.ts` / `async.ts` / `hang.ts` (`hang.ts` is the timeout demo and intentionally exits non-zero).

Generate a declaration file for editor autocomplete:

```
sercon --emit-dts sercon.d.ts
```

Embed in your own Go program:

```
eng := scriptengine.New(scriptengine.Options{
    Timeout:    5 * time.Second,
    ScriptRoot: "./scripts",
})
eng.Register("upper", strings.ToUpper)
val, err := eng.RunFile(ctx, "scripts/main.ts")
```

3. Library API — `pkg/scriptengine`

Engine, Options, New

```
type Options struct {
    Timeout      time.Duration // wall-clock limit per Run; 0 disables
    ScriptRoot   string        // base for require/import resolution
    DisableConsole bool          // turn off the goja_nodejs console module
    Verbose      io.Writer     // diagnostic traces, prefixed "[sercon] "
    ModuleLoader func(candidatePath string) (source string, found bool,
err error)
}

func New(opts Options) *Engine
```

`ModuleLoader`, when non-nil, is consulted for every require/import candidate path **before** the filesystem — the hook for embedders that want to serve modules from somewhere other than disk (an in-memory FS, a network source, an embedded bundle). goja probes several candidates per specifier (`./x`, `./x.ts`, `./x/index.ts`, ...); the engine also tries the bare path plus the usual extension fallbacks (`.ts` / `.tsx` / `.js` / `.mjs` / `.cjs` / `.json`) against the loader, so a loader can match on a plain suffix. Return `(source, true, nil)` to serve the module (transpiled when the path ends in `.ts` / `.tsx`, exactly like a disk read); `("", false, nil)` to fall through to the filesystem; `("", false, err)` to abort resolution.

```
eng := scriptengine.New(scriptengine.Options{
    ModuleLoader: func(path string) (string, bool, error) {
        if strings.HasSuffix(path, "greeting.ts") {
            return `export const hi = (n: string) => "hi " + n;`, true, nil
        }
        return "", false, nil // fall through to disk
    },
})
```

`Engine` is the host of all registered bindings. Construct it once, then call `Run` / `RunFile` as many times as needed — each call gets a *fresh* `*goja.Runtime` so script state never leaks between invocations.

`ScriptRoot` defaults to the working directory at `New` time if left empty. Both `Timeout` and `ScriptRoot` can be overridden on a per-Run basis only by constructing a fresh `Engine`; see [Out-of-scope](#) for the per-Run override on the roadmap.

Registering bindings

Function	Use when...
<code>Register(name, value)</code>	The binding is a pure function, struct, or primitive that doesn't need access to the runtime.
<code>RegisterNamespace(name, members map[string]any)</code>	You want to expose <code>name.x</code> , <code>name.y</code> without defining a Go struct.
<code>RegisterConstructor(name, ctor)</code>	The binding is a Go function whose return value should be treated as a class in the emitted <code>.d.ts</code> . (Runtime semantics today are identical to <code>Register</code> ; the <code>d.ts</code> emitter is the only differentiator.)
<code>RegisterFactory(name, factory func(vm, loop) any)</code>	The binding needs the per-Run <code>*goja.Runtime</code> or <code>*eventloop.EventLoop</code> in scope (typical for Promise-returning I/O).

Function	Use when...
<code>RegisterNamespaceFactory(name, factory)</code>	Same, but the factory returns the full member map.

Example — synchronous function binding:

```
eng.Register("greet", func(name string) string { return "hi " + name })
```

Example — namespace with a sync member:

```
eng.RegisterNamespace("math2", map[string]any{
    "square": func(n float64) float64 { return n * n },
    "pi":     3.14159,
})
```

Example — async (Promise-returning) binding via `PromisifyAsync`:

```
eng.RegisterFactory("httpGet", func(vm *goja.Runtime, loop
*eventloop.EventLoop) any {
    return scriptengine.PromisifyAsync(vm, loop,
        func(ctx context.Context, call goja.FunctionCall) (map[string]any,
error) {
            url := call.Argument(0).String()
            resp, err := http.Get(url)
            if err != nil {
                return nil, err
            }
            defer resp.Body.Close()
            body, _ := io.ReadAll(resp.Body)
            return map[string]any{"status": resp.StatusCode, "body":
string(body)}, nil
        })
})
```

Run, RunFile

```
func (e *Engine) Run(ctx context.Context, name, source string, opts
...RunOption) (goja.Value, error)
func (e *Engine) RunFile(ctx context.Context, path string, opts
...RunOption) (goja.Value, error)
```

`Run` executes `source` as an entry-script TS. `name` is used in stack traces. The returned value is currently always `undefined` — top-level expression capture is on the backlog.

Both methods respect `Options.Timeout` and `ctx` cancellation, whichever fires first; the resulting error is either `ErrScriptTimeout`, `ctx.Err()`, or the underlying JS exception.

Per-Run options

```
type RunOption func(*runConfig)

func WithScriptRoot(dir string) RunOption
```

Reuse a single Engine across many scripts that each live in their own directory by passing `WithScriptRoot(dir)` to `Run` / `RunFile`. The override applies only to this call; `Options.ScriptRoot` is used for every other `Run`.

```
_, _ = eng.Run(ctx, "main.ts", source,
scriptengine.WithScriptRoot("/path/to/run42"))
```

```
func WithArgs(args []string) RunOption
```

Set the per-script argument vector. The script reads it as `runtime.argv`, a global with Node/Bun layout: `argv[0]` is `Options.ProgramName` (defaults to `filepath.Base(os.Args[0])`); the CLI sets it to `"sercon"`, `argv[1]` is the running script's path, and the `WithArgs` values follow from index 2. The global is always present — with no args it is just `[programName, scriptPath]`.

```
_, _ = eng.Run(ctx, "main.ts", source, scriptengine.WithArgs([]string{"--
port", "8080"}))
// inside the script: runtime.argv === [ProgramName, "/abs/main.ts", "--
port", "8080"]
```

Reset

```
func (e *Engine) Reset()
```

Clears every registered binding. Useful when a long-lived Engine is reused across unrelated batches of scripts that each want a clean global namespace. **Not safe to call concurrently with `Run` / `RunFile`.**

WriteTypes

```
func (e *Engine) WriteTypes(w io.Writer) error
```

Emits a `.d.ts` describing the registered surface. See section [14. Type generation](#) for what the mapping looks like.

SetDocs and SetMemberDocs

```
func (e *Engine) SetDocs(path, doc string)
func (e *Engine) SetMemberDocs(namespace string, docs map[string]string)
```

Attach JSDoc strings to registered bindings so the emitted `.d.ts` grows readable editor hover. `SetDocs` takes a dotted path — either a bare top-level name (`"greet"`, `"http"`) or `"namespace.member"` for namespace members (`"http.get"`, `"exec.shell"`); `SetMemberDocs` is the convenience for documenting many members of a namespace at once (the namespace itself can still be documented separately via `SetDocs`).

Multi-line docs are written as `\n`-separated strings; the emitter expands them into a standard `*`-prefixed JSDoc block. Single-line docs collapse to `/** ... */`. Calling `SetDocs` with an empty string removes any previous doc for that path. Bindings without a doc entry get no JSDoc block at all — the emitter doesn't insert empty `/** */` placeholders.

`SetDocs` may be called before or after the matching `Register...` call; docs are looked up by name at emit time. Like the registration methods, `SetDocs` must not race with a `Run` / `WriteTypes` / `Reset` on the same engine.

```
eng.Register("greet", func(name string) string { return "hi " + name })
eng.SetDocs("greet", "Greet someone by name.")

eng.RegisterNamespace("math2", map[string]any{
    "pi":      3.14159,
    "squared": func(n float64) float64 { return n * n },
})
eng.SetDocs("math2", "Tiny math helpers.")
eng.SetMemberDocs("math2", map[string]string{
    "pi":      "Circumference / diameter ratio.",
    "squared": "Multiply a number by itself.",
})
```

PromisifyAsync[T] and AsyncBinding

```
type AsyncBinding struct {
    Func          func(goja.FunctionCall) goja.Value
    TSReturnType string
}

func PromisifyAsync[T any](vm *goja.Runtime, loop *eventloop.EventLoop,
    work func(ctx context.Context, call goja.FunctionCall) (T, error),
) AsyncBinding
```

`PromisifyAsync` turns blocking Go work into a JS Promise. It launches the work in a goroutine, parks a `SetTimeout` sentinel so `loop.Run` doesn't return early, and schedules the

resolve/reject back onto the event loop. **Required** for any Promise-returning binding — `RunOnLoop` alone is not counted as a live job by the event loop.

`PromisifyAsync` returns an `AsyncBinding` carrier rather than the bare callback. The engine unwraps it to `.Func` at `vm.Set` time so goja's special-case detection of `func(goja.FunctionCall) goja.Value` host-callbacks still fires; the `.d.ts` emitter reads `.TSReturnType` to emit `Promise<T>` (mapped from the generic `T` via `reflect` at construction time) instead of the previous `unknown`. Hosts assigning the result into a `map[string]any` for `RegisterNamespace` / `RegisterNamespaceFactory` get the unwrap recursively too — no manual work required.

Version

```
import "github.com/codedeviate/sercon/pkg/scriptengine"

fmt.Println(scriptengine.Version) // "0.1.0"
```

Bumped in lockstep with the git tag.

4. CLI — `sercon`

```
sercon [flags] <script.ts> [script.ts ...]
sercon [flags] - # read entry script from stdin
sercon --examples | --help | --version
```

Flag	Purpose
<code>-timeout DURATION</code>	Wall-clock limit per script (default <code>10s</code> ; <code>0</code> disables).
<code>-root DIR</code>	Override the script root for <code>require</code> / <code>import</code> resolution.
<code>-emit-dts PATH</code>	Write the example bindings' <code>.d.ts</code> to <code>PATH</code> and exit.
<code>-v</code>	Verbose: trace the rewritten entry-script JS and each module resolution to stderr; also print duration on failure.
<code>-h</code> , <code>--help</code>	In-depth colored help.
<code>--examples</code>	In-depth colored walkthrough of every feature.
<code>--version</code>	Print the engine version (plus goja/esbuild build-info versions).
<code>--watch</code>	Re-run on file change under the script root. Debounced 150 ms. Ctrl-C exits.

Each positional argument is either a path to a `.ts` / `.tsx` file or `-` to read an entry script from standard input:

```
echo 'runtime.log(1 + 2);' | sercon -  
sercon prelude.ts -                # prelude then stdin
```

Passing arguments: `--` and `runtime.argv`

Everything after a standalone `--` is the script argument vector, exposed to every script via `runtime.argv`:

```
sercon run.ts -- --port 8080 hello
```

`runtime.argv` uses the Node/Bun layout `[programName, scriptPath, ...userArgs]` — `argv[0]` is `"sercon"`, `argv[1]` is the path of the script currently executing, and the post-`--` arguments start at index 2 (so `runtime.argv.slice(2)` is just your args). All scripts in one invocation share the same `--` tail, but each sees its own path at `argv[1]`. The global is always present; with no `--` it is just `[programName, scriptPath]`.

```
const args = runtime.argv.slice(2); // ["--port", "8080", "hello"]
```

Exit codes are distinct per failure type so shells can react sensibly. When several scripts run, the highest applicable code wins:

Code	Meaning
0	every script passed.
1	CLI usage error (unknown flag, missing script argument, ...).
2	at least one script failed to transpile — never ran.
3	at least one script timed out (<code>-timeout</code>) or was context-cancelled.
4	at least one script ran and threw a JS exception.

`-v` writes lines prefixed with `[sercon]` to stderr. The traces include the full rewritten entry-script JS (the form goja actually runs, after the ESM→CJS rewrite + async IIFE wrapper) and every module-resolution event, so debugging an unexpected resolve target or a mis-rewritten import is straightforward.

The CLI registers ten reserved top-level globals; see the next section.

`--watch`: re-run on file change

`sercon --watch <script.ts>` runs the script once and then blocks, re-running it every time a watched file changes under the script root. Useful for iterating on a script body or on the

binding surface during development.

```
sercon --watch examples/scripts/smoke.ts
```

What's watched:

- The script root (resolved the same way as the one-shot mode — `-root DIR` if set, else `dirname` of the first script argument).
- Recursively. Symlinks aren't followed.
- New directories that appear during the session are picked up automatically.

What's filtered:

- **File extensions:** `.ts` / `.tsx` / `.js` / `.jsx` / `.json` trigger a re-run; `.d.ts` files match via suffix too. Anything else (Markdown, images, editor swap files, build artefacts) is ignored.
- **Directories:** hidden directories (`.git`, `.vscode`, anything starting with `.`) and `node_modules` are excluded — both because they generate floods of irrelevant events and because recursing into them inflates the watcher's directory count for no gain.

Re-runs are **debounced 150 ms**. Editors typically fire several events per save (write → rename → chmod); collapsing them into a single re-run keeps the loop responsive without thrashing. The debounce window resets on every event, so a rapid burst of saves becomes one re-run after the burst settles.

Each run is delimited by a line like `--- sercon re-run @ HH:MM:SS ---` so the output is visually distinct from the previous run.

`Ctrl-C` (SIGINT) or `SIGTERM` exit cleanly. Per-script failures inside a watch session log as `FAIL` but don't terminate the loop — a syntax error you're trying to fix shouldn't kick you out of watch mode.

The watcher reuses the same `Engine` across runs. Each `Run` already gets a fresh `*goja.Runtime`, so re-running is just re-invoking `runOne` on the existing engine — there's no per-iteration setup cost beyond the transpile + module-resolution work.

Module-graph invalidation. Watch mode tracks each entry script's import graph (its own file plus every module it resolves, captured via `Engine.SetResolveHook`) during the run. On a file change it re-runs **only the entries whose graph includes the changed file** — editing a helper that just one of three entry scripts imports re-runs that one entry, not all three. An entry's own file counts (editing it re-runs it); stdin entries and any entry whose graph isn't known yet re-run unconditionally (conservative). Before each re-run the module cache is busted (`Engine.ResetModuleCache`) so an edited import's new source actually takes effect — the registry otherwise caches compiled bytecode across runs.

Shebang lines and executable scripts (`sercon run`)

A script may begin with a `#!/` shebang line. It is stripped before transpile (the line is blanked in place, so transpile/syntax error line numbers still match the source), which means a `.ts / .tsx / .js` file can be made directly executable:

```
#!/usr/bin/env sercon
runtime.log("hello from an executable script");
```

```
chmod +x hello.ts
./hello.ts
```

The kernel launches a shebang script as `sercon <script> <args...>`, and in the default mode every positional is treated as a *separate script* — so positional arguments to a shebang script would be mistaken for additional script paths. For an executable script that takes arguments, use the **run subcommand** in the shebang:

```
sercon run [flags] <script.ts> [args...]
```

```
#!/usr/bin/env -S sercon run
runtime.log("args:", JSON.stringify(runtime.argv.slice(2)));
```

`sercon run` executes exactly one script and hands every token after the script path to it as `runtime.argv[2:]` (Node/Bun layout: `[program, script, ...args]`) — no standalone `-` separator is needed (and a shebang line can't inject one). The `env -S` form splits the single shebang argument so the kernel runs `sercon run /abs/path/script.ts arg1 arg2 ...`. Flags (`-timeout`, `-root`, `-v`) are accepted before the script path; everything from the script path onward is positional and becomes script args. `env -S` is supported by GNU coreutils, macOS, and the BSDs.

```
sercon run script.ts --port 8080 alice    # argv[2:] = ["--port", "8080", "alice"]
```

sercon serve: long-running scripts with production niceties

```
sercon serve [flags] <script.ts> [-- args...]
```

`sercon serve` is a sibling subcommand that wraps the same engine but adds the conveniences a process supervisor (systemd, docker, launchd) usually wants from a long-lived script. Use it whenever a script binds an HTTP/HTTPS listener (or any future `server.*` protocol) and is expected to keep running. Vanilla `sercon script.ts` also keeps the loop alive while listeners are bound, but gives you none of the deltas below.

Flag	Purpose
<code>--shutdown-timeout DURATION</code>	Graceful-shutdown deadline on SIGTERM/SIGINT (default <code>30s</code>). After the deadline the engine hard-cancels.
<code>--port-override N</code>	If non-zero, replaces every literal <code>port:</code> in <code>server.*.listen({...})</code> calls with <code>N</code> . Useful for spinning up the same script on a different port without editing source. Only literal port values are rewritten — computed expressions (<code>{port: getPort()}</code>) are left alone.
<code>-timeout DURATION</code>	Per-script wall-clock limit. Defaults to <code>0</code> (disabled) under <code>serve</code> so listeners don't get cut off; opt back in if you want one.
<code>-root DIR</code>	Script root for <code>require / import</code> resolution (same semantics as vanilla <code>sercon</code>).
<code>-v</code>	Verbose engine tracing to stderr.

Behavioural deltas vs vanilla `sercon` :

- **Access log to stderr.** Each request emits one line in the format `ts remote method path status dur_μs`, e.g. `2026-05-29T10:23:14Z 127.0.0.1:54321 GET /users/42 200 1843μs`. WebSocket upgrades log only the upgrade line; per-frame traffic is suppressed.
- **Readiness signal to stdout.** When each listener calls back into the binding with `bound`, the binding prints `READY` listening on `tcp/0.0.0.0:8080` to stdout. Multiple listeners produce one `READY` line each. Supervisors that read-line on startup (`systemd-notify`, `docker-compose` healthchecks) can wait on it.
- **Graceful shutdown.** SIGTERM / SIGINT triggers `srv.close()` on every active listener concurrently, then waits up to `--shutdown-timeout` for the script to drain. Clean exit is exit code `0`. Past the deadline, `loop.Terminate()` fires and the process exits with the usual classification.
- **No `--watch`.** Re-running while a listener is bound would race port rebinding; `sercon serve --watch ...` exits with a usage error. Use vanilla `sercon --watch script.ts` for the hot-reload development loop.

```
sercon serve examples/scripts/server-http.ts
sercon serve --port-override 9090 server.ts
sercon serve --shutdown-timeout 10s server.ts
```

5. Reserved globals (script surface)

`sercon` scripts get ten reserved top-level globals. Use them directly — there is no enclosing namespace. The full per-binding reference is the generated `examples/scripts/sercon.d.ts`; this section is the at-a-glance prose.

Reserved names: `runtime`, `crypto`, `text`, `codec`, `fs`, `net`, `db`, `server`, `services`, `tui`. User code can shadow these with a local `let` / `const` / `var` per normal JavaScript scoping — sercon does not intervene at runtime. `server` (added in v0.10.0) covers inbound listeners — see [section 6](#) for the long-form treatment.

Migration from v0.8.0

v0.8.0	v0.9.0
<code>api.runtime.log</code>	<code>runtime.log</code>
<code>api.crypto.hash.sha256</code>	<code>crypto.hash.sha256</code>
<code>api.text.str.trim</code>	<code>text.str.trim</code>
<code>api.format.barcode.encode</code>	<code>codec.barcode.encode</code>
<code>api.fs.path.basename</code>	<code>fs.path.basename</code>
<code>api.net.http.get</code>	<code>net.http.get</code>
<code>api.db.sqlite.open</code>	<code>db.sqlite.open</code>
<code>api.tools.exec.shell</code>	<code>services.exec.shell</code>
<code>api.tools.git.commit</code>	<code>services.git.commit</code>
<code>api.ui.tui.layout</code>	<code>tui.layout</code>
<code>Sercon.argv</code>	<code>runtime.argv</code>

The `api` global no longer exists; reading it throws a `ReferenceError`. The `Sercon` global is also gone.

A mechanical `sed` pass over your scripts handles the prefix drop and the three renames — see the `CHANGELOG.md` [0.9.0] "Changed" entry for a sketch.

`runtime`

Script-host scaffolding. Members:

- `runtime.log(...args)` — stringify each argument and `println` the result.
- `runtime.assert.equal(actual, expected, msg?)` / `runtime.assert.ok(cond, msg?)` — throw on mismatch / falsy.
- `runtime.time.nowMs()` — current wall-clock in milliseconds.
- `runtime.time.sleep(ms)` — Promise that resolves after `ms` on the event loop.
- `runtime.time.format(unixMs, layout, tz?)` — Go-style time layout (e.g. "2006-01-02 15:04:05").
- `runtime.env.get(name)` — process environment variable; returns `undefined` when unset (never throws).

- `runtime.argv` — `[programName, scriptPath, ...userArgs]`. User args come from the CLI args after a `--` separator. The layout mirrors Node/Bun: `argv[0]` is `"sercon"`, `argv[1]` is the path of the currently-executing script (so each script in a multi-arg invocation sees its own path), and `argv.slice(2)` is just your args. The property is always present; with no `--` it is just `[programName, scriptPath]`.

```
runtime.log("hello");
runtime.assert.equal(2 + 2, 4);
const args = runtime.argv.slice(2); // post-`--` user args
```

crypto

Hashing, JWT, and age-style encryption. Members:

- `crypto.hash.*` — `md5`, `sha1`, `sha256`, `sha384`, `sha512`, `sha3_256`, `sha3_512`, `blake3`, `crc32`. All take a `string` and return the hex digest.
- `crypto.jwt.sign(claims, key, opts?)` / `crypto.jwt.view(token)` / `crypto.jwt.validate(token, key, opts?)` — HS256/RS256/ES256 JWTs.
- `crypto.encrypt.{keygen, keygenPgp, encrypt, decrypt, rekey, detectBackend}` — age-format symmetric encryption with PGP-compatible key wrapping.

text

String, regex, charset, and data manipulation. Members:

- `text.str.*` — `trim`, `ltrim`, `rtrim`, `reverse`, `stripHtml`, `nl2br`, `br2nl`, `base64Encode`, `base64Decode`, `urlEncode`, `urlDecode`, `htmlEntityDecode`, `pad/lpad/rpad`, `sprintf`, `printf`, `normalizeNewlines`.
- `text.preg.*` and `text.preg2.*` — PCRE-style regex (PHP-compatible named-capture, lookbehind, etc.).
- `text.charset.detect(data)` / `decode(data, charset)` / `encode(text, charset)` — character-set detection and conversion.
- `text.jq.query(json, expr)` / `queryAll(json, expr)` — jq filtering against parsed JSON values.
- `text.diff.compare(a, b, opts?)` — unified / patience / inline diffs between two strings.

codec

Binary-format codecs (was `format` in v0.8.0). Members:

- `codec.compression.{algos, compress, decompress}` — gzip, deflate, zlib, bzip2, zstd, brotli, lz4, xz, snappy.
- `codec.barcode.{formats, decodableFormats, encode, decode}` — QR, DataMatrix, Aztec, PDF417, Code128, Code39, Codabar, EAN-13/8, UPC-A, ITF. `encode`

returns PNG bytes; `decode` reads image bytes.

- `codec.checkdigit.{algos, validate, compute, inspect}` — Luhn, ISBN-10/13, EAN-13/8, UPC-A.
- `codec.php.{serialize, unserialize, varExport, parseVarExport, varDump, parseVarDump}` — read and write PHP data dumps.
- `codec.perl.{dumper, parseDumper}` — read and write Perl `Data::Dumper` output.

`codec.php` — PHP dump formats

Three PHP textual formats, each with an encoder and a matching parser:

- `codec.php.serialize(value, opts?): string` — PHP `serialize()`. Emits the canonical wire form (`s:`, `i:`, `d:`, `b:`, `N;`, `a:`, `0:`).
- `codec.php.unserialize(text, opts?): value` — inverse of `serialize`.
- `codec.php.varExport(value, opts?): string` — PHP `var_export()`: valid PHP source (`array ( ... )`, `\Cls::__set_state( ... )`).
- `codec.php.parseVarExport(text, opts?): value` — read a `var_export` literal back to a value.
- `codec.php.varDump(value, opts?): string` — PHP `var_dump()`: human-readable debug output with type/length annotations.
- `codec.php.parseVarDump(text, opts?): value` — **best-effort** read of `var_dump` output.

Array heuristic. A JS array, or a JS object whose keys are exactly the contiguous integer sequence `0..n-1`, maps to a PHP list array. Any other object maps to a PHP associative array (string/int keys preserved).

The `__class` sentinel. An object carrying a `__class` string key (configurable via `opts.classKey`, default `"__class"`) round-trips as a PHP *object* — `serialize` emits `0:...`, `var_export` emits `\Cls::__set_state(...)`, `var_dump` emits `object(Cls)#...`. The remaining keys become the object's properties. Decoding restores the `__class` key. Without the sentinel a value is treated as an array.

Shared references and cycles. `serialize/unserialize` model PHP's `r:` / `R:` reference markers: when the same object appears more than once, the decoder resolves both occurrences to the *same* JS object (a DAG is preserved). A true cycle (an object reachable from itself) **throws** on encode — there is no JS value that can represent it losslessly here.

`var_dump` parse is lossy. PHP's `var_dump` is a debug view, not a serialization format; `parseVarDump` is best-effort and **throws** when it hits something it cannot faithfully reconstruct: the `*RECURSION*` marker, a length-truncated string (its declared byte length disagrees with the bytes present), or visibility-annotated property names (`["x":"Cls":private]`).

`codec.perl` — Perl `Data::Dumper`

- `codec.perl.dumper(value, opts?): string` — emit a `Data::Dumper`-style dump (`$VAR1 = ... ;`) with normalized indentation.

- `codec.perl.parseDumper(text, opts?): value` — read `Data::Dumper` output back to a value.

The JSON bool convention. Perl has no native boolean, so JSON modules represent one as a blessed scalar reference (e.g. `bless( do{\(my $o = 1)}, 'JSON::XS::Boolean' )`). `dumper` emits JS booleans in this form, and `parseDumper` decodes a blessed scalar ref in the JSON-bool family (`JSON::XS::Boolean`, `JSON::PP::Boolean`, ...) back to a JS boolean. A bare `1 / 0` stays a number. The class used on encode is `opts.perlBoolClass` (default `"JSON::XS::Boolean"`). Blessed hashes carry the `__class` sentinel, mirroring the PHP side. Cycles **throw**.

`opts` and shared caveats

All `codec.php.*` / `codec.perl.*` functions accept an optional `opts` bag:

- `classKey` (default `"__class"`) — the key used as the class sentinel.
- `perlBoolClass` (default `"JSON::XS::Boolean"`) — class for Perl booleans.
- `indent` — override the default 2-space indentation step (`varExport`, `dumper`).

Caveats shared with JSON: strings are UTF-8 and lengths are reported in **bytes** (multibyte chars count as more than one); JS has a single number type, so `int` and `float` collapse the same way `JSON.stringify` does (e.g. `1.0` may re-emit as `1`). Decoded objects preserve a stable key order, so re-encoding is byte-stable — safe for canonical-JSON / payment hashing.

`fs`

Filesystem operations:

- `fs.path.dirname(p)` / `fs.path.basename(p, suffix?)` — POSIX-style.
- `fs.archive.create(srcPath, opts?)` / `fs.archive.extract(archivePath, destDir, opts?)` — tar/zip with optional compression.

`net`

Network clients and probes:

- `net.http.{get, post, request}` — fetch-style HTTP client; `request` accepts headers, body, timeout, retry, follow, and basic auth.
- `net.probe.{tcp, dns, tls, ntp, whois, ping, smtp, wss}` — one-shot reachability / handshake probes. Each returns a structured result; failures surface as `Error` objects with details.
- `net.netstatus.check(host)` — combined reachability summary.
- `net.email.*` — `spf`, `dmARC`, `mtaSTS`, `tlsRpt`, `bIMI`, and `all(domain)` which runs every probe in parallel and aggregates; plus `send({...})` — outbound SMTP sender (see §6.7).
- `net.browser.open(url)` — launches the OS default browser.

db

Database / KV / directory clients:

- `db.sqlite.open(path)` — embedded SQLite (cgo-free driver).
- `db.redis.open(addr, opts?)` — Redis client.
- `db.memcached.open(addr)` — Memcached client.
- `db.ldap.open(url, opts?)` — LDAP client (search, bind).
- `db.dict.{define, match}` — local dictionary lookup / fuzzy match.

services

Subprocess and external-CLI / service wrappers (was `tools` in v0.8.0). Members:

- `services.exec.shell(cmd, opts?)` — run a shell command; captures stdout/stderr or streams into a `tui` pane when `opts.pane` is set.
- `services.exec.http(method, url, opts?)` — shell-level `curl`-style HTTP (separate from `net.http`, intended for raw protocol use).
- `services.git.*` — git porcelain wrappers (clone, status, commit, log, ...) invoking the local `git` binary.
- `services.gh.{authStatus, prList, repoView}` — thin `gh` CLI wrappers.
- `services.ai.{providers, send}` — provider-agnostic AI chat client.

tui

Multi-pane terminal UI (was `ui.tui` in v0.8.0). `tui.layout(tree)` declares panes;

`tui.pane(name)` returns a pane handle with `write`, `writeln`, `clear`, `title`.

`services.exec.shell({pane})` streams subprocess I/O into a pane.

Scripts that orchestrate multiple subprocesses (e.g. `brew`, `npm`, `cargo` updates running in parallel) can declare a multi-pane terminal layout and route each subprocess's stdout/stderr into its own pane. Activation is **script-driven**: the first call to `tui.layout(...)` enters TUI mode. If stdout is not a TTY (CI, pipes, `make demo`), the same calls fall back to prefixed plain-text lines (`[paneName] line`) so the same script runs in both contexts.

Layout

```
tui.layout({
  rows: [
    { name: "log", title: "Orchestrator", weight: 1 },
    { cols: [
      { name: "brew" },
      { name: "npm" },
    ],
      weight: 2,
    },
  ],
});
```

Each layout node is one of:

- `{ name: string, title?: string, weight?: number }` — a leaf pane.
- `{ rows: LayoutNode[], weight?: number }` — a vertical split.
- `{ cols: LayoutNode[], weight?: number }` — a horizontal split.

`weight` defaults to 1. Pane names must be unique across the whole tree. `layout` may be called **once** per Run; a second call throws.

Pane handles

```
const log = tui.pane("log");
log.writeln("Updating Homebrew...");
log.title("Done");
log.clear();
log.write("partial line ");
```

`tui.pane(name)` returns a handle with `write`, `writeln`, `clear`, `title`. Methods are synchronous from the script's perspective — they enqueue to the TUI goroutine.

Subprocess routing

`services.exec.shell(cmd, opts)` accepts `opts.pane` (a Pane handle or a pane name string):

```
await services.exec.shell("brew update && brew upgrade", { pane: "brew" });
```

When set, the subprocess's stdout **and** stderr stream into the pane line by line. The returned `{ exitCode, durationMs, success }` is the same as without `pane`: except `stdout` and `stderr` are empty (data was streamed, not captured). ANSI colors are translated to `tview` color tags and rendered natively; `\r` without `\n` overwrites the current line so progress spinners render cleanly.

Keybindings (TTY mode only)

Key	Action
Tab / Shift-Tab	Cycle focus through panes
PgUp / PgDn	Scroll focused pane one page
↑ / ↓	Scroll focused pane one line
Home / End	Jump to top / bottom
Ctrl-C	Abort the script

The focused pane has a yellow border; the bottom status bar shows its name and the active keys.

Limitations (v1)

- `tui` is **incompatible with `--watch`**: calling `tui.layout()` under `--watch` throws. Use one or the other.
- No mouse, no runtime layout mutation, no input panes (`tui.input`), no snapshot-to-normal-screen on exit. The alt screen is restored at script end and the usual `PASS / FAIL` line prints.

6. Servers

The `server` global (added in v0.10.0) hosts inbound listeners — scripts that *accept* connections rather than make them. The first sub-spec ships `server.http` and `server.https`; future cycles will add SMTP, IMAP, FTP, and POP3 against the same engine foundation.

6.1 Foundation: `HoldRun` + `LoopCallable`

Two engine primitives back every long-lived binding. They live on `*scriptengine.Engine` (CLI use of them is internal; the public contract is "use these if you write a similar binding yourself").

`Engine.HoldRun(reason string) (release func())` — keeps `loop.Run` from exiting while a binding has resources outstanding. The event loop normally exits as soon as its `jobCount` drops to zero; `setTimeout` / `setInterval` / `setImmediate` are the only things that count. `HoldRun` parks a 24-hour sentinel timer under the hood (same trick `PromisifyAsync` uses for one-shot async work), returns a `release` function that clears it, and increments a refcount so multiple concurrent holders compose. `release` is idempotent. Any sentinels still parked when `Run` returns are cleanup-drained automatically as a safety net — a binding that forgets to call `release` does not leak into the next `Run`.

`scriptengine.NewLoopCallable(loop, fn)` — wraps a captured `goja.Callable` (a JS function reference) so it can be invoked from *any* goroutine. Goja's runtime is single-threaded;

calling a Callable directly from a non-loop goroutine corrupts engine state.

`LoopCallable.Call(buildArgs)` enqueues a `loop.RunOnLoop` callback, builds the arg list on the loop (where `vm.ToValue` is valid), invokes the callable, and returns the result to the caller's goroutine. `LoopCallable.CallOnLoop(vm, args...)` is the already-on-the-loop variant — using `Call` from inside a loop callback deadlocks because the new `RunOnLoop` job can't run until the current one returns.

You use them together. A typical server binding holds one or two `LoopCallable`s for the user's handlers, parks one `HoldRun` sentinel per active listener, and releases on close. The CLI bindings under `cmd/sercon/api_server*.go` follow this pattern.

Concurrency model. Because every handler invocation marshals back onto the single goja loop, **handlers serialize**: only one JS handler runs at a time, across all listeners. This is a feature (no JS data races, no shared-state confusion) and a throughput ceiling. For CPU-bound work in a handler, offload to a Go binding that does the work in a goroutine and resolves a Promise.

6.2 `server.http.listen` / `server.https.listen`

```
// HTTP
const srv = await server.http.listen({
  port: 8080,                // required
  host: "0.0.0.0",           // default
  routes: { "GET /": (req, res) => res.text("hi") },
  use: [logger],              // optional global middleware
});

// HTTPS – identical plus cert/key (file paths OR inline PEM strings)
const srv2 = await server.https.listen({
  port: 8443,
  cert: "/etc/ssl/server.pem",
  key: "/etc/ssl/server.key",
  routes,
});

srv.address;                // "tcp/0.0.0.0:8080"
await srv.close();          // graceful: stop accepting, drain, release HoldRun
await srv.stopped;          // Promise that resolves when the listener exits
```

Options:

Key	Type	Notes
<code>port</code>	<code>number</code>	Required. Bind port. <code>0</code> lets the kernel pick; read the chosen port off <code>srv.address</code> .

Key	Type	Notes
host	string	Default "0.0.0.0". Pass "127.0.0.1" for loopback-only.
routes	Record<string, RouteValue>	Required (may be {}). Keys are stdlib http.ServeMux patterns.
use	Middleware[]	Optional global middleware; runs in array order before any per-route middleware.
cert	string	HTTPS only. File path or inline PEM.
key	string	HTTPS only. File path or inline PEM.

Route patterns are stdlib Go 1.22+ http.ServeMux syntax: "METHOD /path/{param}/{rest...}" . The leading method is optional (omit to match any). {name} captures one path segment; {name...} captures the tail (wildcard). No new routing library — stdlib does the matching, including longest-prefix wins. Captured values land in req.params .

RouteValue is either a bare handler — (req, res) => res.text(...) — or an object { use?: Middleware[], handler: HandlerFn } for per-route middleware (see §6.4).

Request shape (req):

```
type Request = {
  method:    string;           // "GET"
  url:       string;           // full URL incl. scheme + host
  path:      string;           // "/users/42"
  query:     Record<string, string[]>; // ?a=1&a=2 → {a: ["1","2"]}
  headers:   Record<string, string[]>; // lowercase keys
  params:    Record<string, string>;    // path-pattern captures
  body:      string;            // UTF-8 decoded request body
  bodyBytes: Uint8Array;        // same data as raw bytes
  remote:    string;            // "1.2.3.4:54321"
  cookies:   Record<string, string>;    // parsed Cookie header
};
```

Response builder (res) — every method returns res for chaining; the terminal .json / .text / .html / .bytes / .empty / .redirect sets the body and flips an internal finalized flag:

```

type CookieOpts = {
  domain?: string;
  path?: string;
  maxAge?: number; // seconds; 0 = session; <0 = delete
  expires?: number; // unix ms
  secure?: boolean;
  httpOnly?: boolean;
  sameSite?: "strict" | "lax" | "none";
};

type Response = {
  status(code: number): Response;
  header(name: string, value: string): Response;
  cookie(name: string, value: string, opts?: CookieOpts): Response;
  // Terminals (all return res so they remain chainable but no further
  // write makes sense after one fires; a second terminal throws):
  json(value: unknown): Response; // → application/json
  text(s: string): Response; // → text/plain
  html(s: string): Response; // → text/html
  bytes(b: Uint8Array, ct?: string): Response; // default
  application/octet-stream
  empty(): Response; // no body; pair with .status() for
  204/304
  redirect(loc: string, code?: number): Response; // default 302
  // Upgrade (WebSocket):
  upgradeWebSocket(opts?: { readBuffer?: number }): Promise<WebSocket>;
};

```

After the handler's returned Promise resolves, the engine checks `res.finalized`:

- **true** — buffered status / headers / body get written to the underlying `http.ResponseWriter`.
- **false** — engine sends `204 No Content`.
- **handler threw** (sync or async) — engine sends `500 Internal Server Error` and logs the error (`stderr` under vanilla `sercon`; access log under `sercon serve`). The script keeps running; only that one request is affected.

Calling a terminal twice on the same `res` throws a `TypeError` (the second call sees `finalized === true` and refuses).

Multiple listeners compose naturally. A script can run

```

const [a, b] = await Promise.all([
  server.http.listen({ port: 80, routes }),
  server.https.listen({ port: 443, routes, cert, key }),
]);

```

Each `listen` call holds its own `HoldRun` sentinel; closing one listener does not affect the other.

6.3 Static-file serving

```
"GET /assets/{rest...}": server.http.static({
  dir:          "/var/www/public",    // root directory on disk
  stripPrefix:  "/assets/",           // URL prefix to strip before disk
  lookup
  index?:      "index.html",          // accepted but currently unused
  (stdlib default applies)
  etag?:       true,                  // accepted but currently unused
  (stdlib default applies)
}),
```

`server.http.static({...})` returns an opaque handler the route compiler unwraps and registers directly on the mux. **The route pattern must include a `{rest...}` wildcard** — without it the match only ever produces the literal prefix and no subpath ever resolves on disk. Internally a `stdlib http.FileServer(http.Dir(dir))` wrapped in `http.StripPrefix(stripPrefix, ...)`; ETag and range requests work out of the box. Symlinks outside `dir` are blocked (`stdlib http.Dir`'s default). No directory listing customisation yet — `index` and `etag` options are reserved for future tuning; v0.10.0 inherits the `stdlib` defaults.

6.4 Middleware

Onion model. A middleware is `(req, res, next) => Promise<void>`. `next` is an async function that runs the rest of the chain; awaiting it lets the middleware post-process.

```
const logger = async (req, res, next) => {
  const start = runtime.time.nowMs();
  await next(); // run downstream chain
  runtime.log(req.method, req.path, "->", runtime.time.nowMs() - start,
"ms");
};

await server.http.listen({
  port: 8080,
  use: [logger], // global middleware
  routes: {
    "GET /api/secure": { // per-route middleware
      use: [authCheck],
      handler: (req, res) => res.json({ ok: true }),
    },
  },
});
```

Attachment points:

- **Global** — `use`: array in `listen({...})` options. Runs for every request, in declaration order, before any per-route middleware.
- **Per-route** — when a route value is `{ use: [...], handler: fn }`, those middleware run after globals but before the handler.

For a request to `POST /api/secure` with global middleware `[A, B]` and per-route middleware `[C]`, the composition is:

```
A(req, res, () => B(req, res, () => C(req, res, () => handler(req, res))))
```

Each layer can `await next()` then mutate response headers or record timings, exactly like Express / Koa.

Short-circuit: a middleware that does *not* call `await next()` stops the chain. It must terminate `res` itself (e.g. `res.status(401).text("denied")`) — otherwise the engine sends `204 No Content` per the unfinalized-response rule.

Throwing: any throw mid-chain (sync or via rejected Promise) bubbles to the engine's 500 path. Wrap with `try / catch` if you want custom error handling.

6.5 WebSocket upgrade


```
"GET /ws": async (req, res) => {
  const ws = await res.upgradeWebSocket({ readBuffer: 64 });
  // ws is both an async iterator AND has .send / .close.
  for await (const msg of ws) {
    if (msg.type === "text") {
      await ws.send("echo:" + msg.text);
    } else {
      await ws.send(msg.bytes);          // msg.type === "binary"
    }
  }
  // iterator ends on close; ws.closeCode / ws.closeReason populated
},
```

Type:

```
type WSMMessage =
  | { type: "text";   text: string   }
  | { type: "binary"; bytes: Uint8Array };

type WebSocket = AsyncIterable<WSMMessage> & {
  send(data: string | Uint8Array): Promise<void>;
  close(code?: number, reason?: string): Promise<void>;
  closeCode?: number;
  closeReason?: string;
  remote: string;
};
```

Library: github.com/coder/websocket (the modern successor to nhooyr.io/websocket; [gorilla/websocket](https://gorilla.github.io/websocket) is in maintenance mode). Pure Go, zero deps.

Marshalling. Per upgraded connection, a Go goroutine reads frames into a buffered channel (capacity = `readBuffer`, default 64). Each `next()` on the async iterator pulls one message off the channel and resolves on the loop via `LoopCallable`. `ws.send()` schedules a write back through the connection-owning goroutine.

Backpressure. If the read buffer fills (slow JS consumer), back-pressure propagates into the websocket library's read loop — the client eventually sees `1009 message too big` after the library's own limit (1MB default). Bump `readBuffer` if your workload bursts and you want more in-flight queueing.

`ws.close()` closes the send channel, drains the receive channel, sends a close frame, and ends the async iterator on the next `next()` call. The script's `for await` exits cleanly; the handler's Promise resolves; the connection's goroutine returns. The HTTP server's `HoldRun` is **unaffected** — it stays alive for new connections until `srv.close()`.

`async function*` and `for await (...)` are not natively parsed by goja. esbuild's `Supported` flag lowers both during transpile (see CHANGELOG), and every Run installs

`Symbol.asyncIterator = Symbol.for("@@asyncIterator")` so the lowered helper and user code agree on the same iteration key.

6.6 Lifecycle

Vanilla sercon script.ts. Each `server.*.listen` call parks a `HoldRun` sentinel and the event loop stays alive while the listener is bound. `SIGINT` terminates abruptly: the engine's interrupt watcher calls `loop.Terminate()`, the cleanup drain runs (closing each listener), and the process exits. No access log, no readiness signal, no shutdown timeout.

sercon serve script.ts. Adds the production niceties documented in §4: structured access log to stderr, a `READY` listening on `tcp/...` line on stdout per listener, and graceful shutdown with `--shutdown-timeout` (default `30s`). Clean `SIGTERM` exits `0`. Choose `serve` whenever the script is supervised (systemd, docker compose, launchd).

```
# Quick local test:
sercon examples/scripts/server-http.ts

# Production-style supervised run:
sercon serve examples/scripts/server-http.ts
```

6.7 SMTP

SMTP is a **stateful, multi-stage transaction**: a client connects, optionally authenticates, declares a sender (`MAIL FROM`), one or more recipients (`RCPT TO`), then streams the message body (`DATA`). `sercon` maps each stage to a JavaScript callback so a script can accept or reject the transaction incrementally — refuse an unknown sender at `MAIL`, refuse an unroutable recipient at `RCPT`, or inspect the parsed message at `DATA`. The inbound listener is `server.smtp.listen({...})`; the outbound side is `net.email.send({...})` (covered at the end of this section).

`server.smtp.listen({...})`

Synchronously binds the listening socket (so a port already in use throws immediately), then serves in the background. Returns a handle once the loop schedules it.

```

const srv = await server.smtp.listen({
  port: 2525,
  hostname: "mx.example.com",
  handlers: {
    onMail: (env)      => env.from.endsWith("@example.com"),
    onRcpt: (env, rcpt) => isLocalMailbox(rcpt),
    onData: (env, msg) => { store(msg); return true; },
  },
});

// srv.address  → "tcp/0.0.0.0:2525"
// srv.close()  → Promise<void> (graceful, 30s drain)
// srv.stopped  → Promise<void> (resolves when the listener exits)

```

Options:

Field	Type	Default	Notes
port	number	—	Required. TCP port to bind.
host	string	"0.0.0.0"	Bind address.
hostname	string	os.Hostname()	EHLO greeting + Received: identity.
handlers	{onMail, onRcpt, onData}	—	Required. All three functions; see below.
auth	(user, pass, env) => bool Promise<bool>	—	Optional SASL verifier (PLAIN + LOGIN).
starttls	{cert, key}	—	Enables STARTTLS. File paths or inline PEM.
allowInsecureAuth	boolean	false	Permit AUTH on a plaintext connection (dev only).
maxMessageBytes	number	10485760 (10 MB)	DATA size cap; a non-positive value keeps the default.
maxRecipients	number	100	Max RCPT TO per transaction.

Field	Type	Default	Notes
<code>sessionTimeout</code>	<code>number</code>	<code>30000</code>	Per-session idle timeout, milliseconds.

Handlers. `onMail(envelope)`, `onRcpt(envelope, recipient)`, and `onData(envelope, message)` are invoked at the matching protocol stage. All three are required. Each may be `async` — a returned Promise is awaited before the SMTP response is written. The return value (or a thrown exception) controls the SMTP reply, uniformly across all three:

Return value	SMTP response	Use case
<code>true / undefined</code>	<code>250 OK</code>	Accept the stage
<code>false</code>	<code>550 Command refused</code>	Generic permanent reject
<code>string</code>	<code>550 <string></code>	Permanent reject with a reason
<code>throw</code>	<code>451 Temporary failure</code>	Transient error (client should retry)

Envelope (passed to all three handlers; `recipients` grows as `RCPT TO` accumulates):

```
type Envelope = {
  from:          string;    // "" for the null sender (DSN bounces)
  recipients:    string[];  // accepted RCPT TO so far
  remote:        string;    // "1.2.3.4:54321"
  helo:          string;    // EHLO/HELO hostname the client gave
  authenticatedUser?: string; // set if AUTH succeeded
  tls?:          { version: string; cipher: string }; // set under
STARTTLS
};
```

Message (only `onData`; parsed via `jhillierd/enmime`, with the exact original bytes always available as `raw`):

```

type Message = {
  from:    string;           // From: header (may differ from
  envelope.from)
  to:      string[];         // To: header
  cc:      string[];         // Cc: header
  subject: string;
  headers: Record<string, string[]>; // lowercase keys; multi-valued
  body: {
    text: string;           // text/plain part; "" if absent
    html: string;           // text/html part; "" if absent
  };
  attachments: Array<{
    filename:    string;
    contentType: string;
    bytes:       Uint8Array;
  }>;
  raw: Uint8Array;           // exact DATA bytes (post dot-
  unstuffing)
};

```

Forwarders and DKIM-style signature verifiers should work from `raw`; everyone else uses the parsed fields. `enmime` is lenient with malformed messages, so a script needing strict RFC 5322 conformance can re-parse `raw` itself.

AUTH. Supply an `auth` callback to enable SASL. Both **PLAIN** and **LOGIN** mechanisms are advertised; the callback receives the decoded (`username`, `password`, `envelope`) and returns a boolean (or a Promise of one). By default AUTH is refused on a plaintext connection — the client must complete STARTTLS first. Set `allowInsecureAuth: true` to accept credentials over plaintext; this is a **dev-only** escape hatch for trusted local networks and should never be set in production.

STARTTLS. Pass `starttls: {cert, key}` (file paths or inline PEM, same convention as `server.https`) to advertise STARTTLS. The upgrade happens mid-session on the same connection; once active, `envelope.tls` is populated. The listener itself is plaintext at bind time — there is no implicit-TLS (SMTPS / port 465) inbound mode.

Concurrency. Each connection runs on its own Go goroutine, but every handler invocation **serializes on the goja loop** — exactly one handler runs at a time (the same single-threaded model as the HTTP listener in §6.2). A slow `onData` therefore blocks other connections' handlers. For slow work, acknowledge the stage (`return true`) and process in the background (start a Promise without awaiting it).

A short, self-contained round-trip (bind, send to self, assert receipt):

```

const port = 38095;
let captured: { subject: string; from: string } | null = null;

const srv = await server.smtp.listen({
  port,
  hostname: "test.local",
  handlers: {
    onMail: () => true,
    onRcpt: () => true,
    onData: (env, msg) => {
      captured = { subject: msg.subject, from: env.from };
      return true;
    },
  },
});

await net.email.send({
  to:      "alice@test.local",
  from:    "bob@test.local",
  subject: "round-trip demo",
  body:    "hello from the SMTP demo",
  server:  { host: "127.0.0.1", port, tls: "none" },
});

runtime.assert.equal(captured!.subject, "round-trip demo", "subject");
await srv.close();

```

net.email.send({...}) — outbound

The outbound counterpart lives under `net.email` (next to the `spf` / `dmARC` / ... probes). It opens **one TCP connection per call** — no pooling, no queueing, no automatic retries — composes the MIME body in-tree, and returns a per-recipient outcome.

```

const result = await net.email.send({
  to:      ["alice@example.com", "bob@example.com"], // string OR string[]
  from:    "noreply@my.app",
  subject: "your verification code",
  body:    "your code is 123456",                    // plain text
  html:    "<p>your code is <b>123456</b></p>",        // optional HTML
  alternative
    attachments: [
      { filename: "qr.png", contentType: "image/png", bytes: qrBytes },
    ],
    headers: { "X-App": "myapp" },                    // optional extra
  headers
    server: {
      host: "smtp.example.com",
      port: 587,                                       // default 587
    (submission)
      auth: { username: "u", password: "p" },         // optional PLAIN
    auth
      tls: "starttls",                               // "starttls"
    (default) | "tls" | "none"
      },
      timeout: 30000,                                 // ms; default 30s
    });

// result = { accepted: string[], rejected: Array<{ address, reason }> }

```

MIME layout. text only → text/plain; charset=utf-8 (no multipart). text + html → multipart/alternative. Any attachments → multipart/mixed (each part base64, Content-Disposition: attachment). Date, Message-ID, and MIME-Version are auto-generated; headers are merged on top.

TLS modes. "starttls" (default) connects plaintext then upgrades, refusing AUTH until TLS is active. "tls" is implicit TLS from connect (SMTPS, typically port 465). "none" is plaintext throughout, with AUTH disabled.

Outcomes. Transport-level failures (connection refused, TLS handshake, AUTH, MAIL FROM rejected) **throw** a JS exception. The rejected array captures only individual RCPT TO addresses the server permitted us to attempt but then refused — the transaction continues for the rest, so a partial send still delivers to the accepted recipients.

7. JavaScript runtime built-ins (goja)

scriptengine runs on goja, which implements **ES5.1 + a large subset of ES6+**. The following are present and behave per spec:

Globals

- `Object`, `Array`, `String`, `Number`, `Boolean`, `BigInt`
- `Math`, `Date`, `JSON`, `RegExp`
- `Error`, `TypeError`, `RangeError`, `SyntaxError`, `ReferenceError`, `EvalError`, `URIError`
- `Promise` (provided via goja's native implementation; resolution microtasks are pumped by the event loop)
- `Symbol` (partial: well-known symbols `Symbol.iterator`, `Symbol.asyncIterator`, etc., are supported)
- `Proxy`, `Reflect` (partial)

```
runtime.log(Math.PI.toFixed(4), Math.max(1, 9, 3));
runtime.log(new Date().toISOString());
runtime.log(JSON.stringify({ a: 1, b: [2, 3] }));
runtime.log("abc".repeat(3), "abc".padStart(6, "_"));
```

Collections

- `Map`, `Set`, `WeakMap`, `WeakSet`

```
const counts = new Map<string, number>();
for (const ch of "banana") counts.set(ch, (counts.get(ch) ?? 0) + 1);
runtime.log([...counts]); // [{"b",1}, {"a",3}, {"n",2}]
```

Typed arrays

- `ArrayBuffer`, `DataView`
- `Int8Array`, `Uint8Array`, `Uint8ClampedArray`, `Int16Array`, `Uint16Array`, `Int32Array`, `Uint32Array`, `Float32Array`, `Float64Array`

```
const buf = new ArrayBuffer(4);
new DataView(buf).setUint32(0, 0xCAFEBADE);
runtime.log(new Uint8Array(buf)[0].toString(16)); // ca
```

Iteration and generators

- `for...of`, `for...in`, `spread`, `destructuring`
- Generator functions (`function*`)
- Async generators (`async function*`) — supported via goja's event loop


```
function* fib() {
  let [a, b] = [0, 1];
  while (true) {
    yield a;
    [a, b] = [b, a + b];
  }
}
const it = fib();
const first10 = Array.from({ length: 10 }, () => it.next().value);
runtime.log(first10);
```

Not (yet) provided

- `fetch` — use `net.http.*` from the CLI, or register your own binding.
- `Worker`, threading primitives.
- DOM globals (`window`, `document`).
- Native ES modules (`import` / `export` are *transpiled* to CommonJS at load time — see [section 9](#)).

8. Async runtime additions (goja_nodejs)

The event loop bundles a small Node-compatible runtime on top of goja:

Timers

```
setTimeout(() => runtime.log("after 50ms"), 50);
setInterval(() => runtime.log("tick"), 100);
setImmediate(() => runtime.log("next tick"));
clearTimeout(t); clearInterval(i); clearImmediate(im);
```

The event loop holds the script alive while *any* timer is pending. If your script ends with only `setTimeout` callbacks scheduled, the run will wait for them to fire before returning.

console

```
console.log("info");
console.error("oops");
console.warn("careful");
console.info("fyi");
console.debug("trace");
```

Disable with `Options.DisableConsole = true` if you want a clean global namespace.

`require`

CommonJS `require`, with TS-aware extension fallback added by `sercon` (see [section 11](#)):

```
const helpers = require("./helpers");
helpers.doThing();
```

Both `require("./foo")` and `import { x } from "./foo"` resolve to the same module instance per Run; esbuild rewrites `import` to `require` at transpile time.

9. TypeScript support

TS is transpiled by esbuild in-process. There is no `tsc`, no `tsconfig.json`, no type-checking — types are erased and the resulting JS is what executes. Practical implications:

- All TS syntax esbuild supports is allowed: type annotations, `interface`, `type`, generics, enums (compiled to objects), namespaces, decorators (legacy & TC39 stage 3).
- `import type { X } from "y"` is stripped entirely (no runtime require).
- TS errors are *not* surfaced as build errors — only esbuild parse errors are. Run a separate `tsc --noEmit` in CI if you want type enforcement.
- `tsconfig.json` is **not** consulted.

`.tsx` and `.jsx` are supported via esbuild's `LoaderTSX` / `LoaderJSX`, including for required modules resolved by the engine's extension fallback. JSX has no React runtime in scope by default — either set the factory at the file level with an `@jsx` pragma:

```
/** @jsx h */
function h(tag: string, props: any, ...children: any[]) {
  return { tag, props: props ?? {}, children };
}
export const Box = (label: string) => <div className="box">{label}</div>;
```

...or provide a `React.createElement`-compatible binding from Go and rely on esbuild's default pragma. ESM `export default <value>` also works through the entry-script rewriter's `__esModule ? .default : m` unwrap, so `import answer from "./mod"` resolves to the default export.

10. Top-level `await`

Top-level `await` works in entry scripts:

```
const r = await net.http.get("https://example.com");
runtime.log(r.status);
```

How it works under the hood: esbuild rejects top-level await with the `cjs` output format, so the engine transpiles the *entry* script with `format=esm`, then rewrites `import` statements to `require()` calls and wraps the rest of the body in:

```
;(async () => { /* your transpiled body */ })().then(__resolve, __reject);
```

`__resolve` and `__reject` are bindings the engine sets on the runtime to capture the final settlement. *Required-module* code (anything loaded through `require`) is transpiled with `format=cjs` and does **not** support top-level await — wrap in `(async () => ... )()` yourself if you need it inside a module.

11. Module resolution

The engine plugs a custom source loader into `goja_nodejs/require` that adds:

1. **Direct path:** if the requested file exists on disk, use it. `.ts` / `.tsx` files are transpiled on the fly.
2. **Extension fallback** when the request has no extension: try `.ts`, `.tsx`, `.js`, `.cjs`, `.mjs`, `.json` in that order.
3. **`.js` → `.ts` swap:** if a path ending in `.js` / `.cjs` / `.mjs` is requested but the literal file doesn't exist, the same path ending in `.ts` (or `.tsx`) is tried. Handles `package.json` `main` fields that point at compiled output where only the TypeScript source is on disk.
4. **`package.json` source preference:** when reading a `package.json`, if a `source` field is present and points at an existing `.ts` / `.tsx` file, `main` is rewritten to that path before the registry consumes it. Matches the convention used by parcel, microbundle, and similar bundlers to mark the TS source of truth.
5. **Directory index:** if the resolved path is a directory, try `index.ts`, `index.tsx`, `index.js`.

The base directory follows Node's CommonJS rules — relative imports resolve against the requiring module's directory, courtesy of `goja_nodejs`.

```
// All resolve to ./helpers/assert.ts:
import { check } from "./helpers/assert";
import { check } from "./helpers/assert.ts";
const { check } = require("./helpers/assert");
const { check } = require("./helpers/assert.js");
```

JSON imports work out of the box via either the ESM-style default import (esbuild rewrites it to a `require`) or a direct `require`:

```
import data from "./data.json";
const r = require("./data.json");
```

`node_modules` lookup *works* via the upstream registry, but the source loader doesn't currently transpile through that path — keep TS helpers under `ScriptRoot`.

12. Timeouts and cancellation

Two mechanisms interrupt a running script:

- `Options.Timeout` — wall-clock limit per `Run`. Exceeding it returns `scriptengine.ErrScriptTimeout`.
- `ctx` passed to `Run / RunFile` — cancelling the context returns `ctx.Err()` (typically `context.Canceled` or `context.DeadlineExceeded`).

Both call `vm.Interrupt(...)` and `loop.Terminate()` from a watcher goroutine. This works for **sync** JS — including `while(true){}` — because the goja VM checks the interrupt flag between bytecode steps. A host Go binding that's blocked in cgo or a long syscall isn't interruptible mid-call; if you write such a binding, plumb the engine `ctx` through to it yourself.

```
eng := scriptengine.New(scriptengine.Options{Timeout: 200 *
time.Millisecond})
_, err := eng.Run(ctx, "loop.ts", `while (true) {}`)
// err is scriptengine.ErrScriptTimeout, returned ~200-300ms after Run.
```

13. Error semantics

- A Go binding returning `(T, error)` automatically throws as a JS exception when `err != nil`. The exception's `.message` is `err.Error()`.

```
try { mayFail(); } catch (e) { runtime.log(String(e)); }
```

- A Go binding that `panic`s with a `*goja.Object` (e.g. `vm.NewGoError(...)`) does the same.
- A script throwing a JS error out of the top level surfaces from `Run` as a Go `error` whose message is the JS error's `.message`.
- Timeout errors satisfy `errors.Is(err, scriptengine.ErrScriptTimeout)`. Cancellation errors satisfy `errors.Is(err, context.Canceled)` or `context.DeadlineExceeded`.

14. Type generation (.d.ts)

`Engine.WriteTypes(w)` walks the registered bindings and emits a TypeScript declaration file. The mapping table (abbreviated):

Go type	TS type
<code>string</code>	<code>string</code>
<code>any numeric / float64</code>	<code>number</code>
<code>bool</code>	<code>boolean</code>
<code>[]T</code>	<code>T[] ; []byte → Uint8Array</code>
<code>map[string]T</code>	<code>Record<string, T></code>
<code>*T</code>	<code>T</code> (transparent unwrap)
<code>error</code> (single return)	omitted (collapses to <code>void</code>)
<code>(T, error)</code> (tuple return)	<code>T</code>
<code>interface{}</code> / <code>any</code>	<code>unknown</code>
<code>goja.FunctionCall</code> parameter	folded into <code>(...args: unknown[])</code>
<code>goja.Value</code> return	<code>unknown</code>
<code>scriptengine.AsyncBinding</code> value (from <code>PromisifyAsync[T]</code>)	<code>Promise<T></code>
self-referential types	<code>unknown</code> (cycle detection bails after depth 4)

```
sercon --emit-dts sercon.d.ts
```

In your editor, place the resulting `.d.ts` next to your scripts (or reference it via `/// <reference path="..." />`) for autocomplete.

JSDoc comments on declarations

The emitter pulls JSDoc strings from the engine's doc map (set via `Engine.SetDocs` / `Engine.SetMemberDocs`) and renders them above each declaration. Single-line docs collapse to `/** ... */`; multi-line docs expand to a standard `*`-prefixed block. Bindings without a doc entry emit no JSDoc — the emitter never inserts empty placeholders. Docs are looked up by the same dotted path the binding was registered under, so namespace members get hover text too:

```
// AUTO-GENERATED by scriptengine.Engine.WriteTypes – do not edit by hand.

/** Hashing primitives. */
declare const crypto: {
  hash: {
    /** SHA-256 hex digest of a UTF-8 input. */
    sha256(...args: unknown[]): string;
    /** BLAKE3 hex digest (32-byte output, lukechampine.com/blake3). */
    blake3(...args: unknown[]): string;
    // ...
  };
  // ...
};
```

The CLI's reserved-global surface ships with a curated doc map; see `cmd/sercon/docs.go` for the source of truth and add new entries there when adding new bindings (the lockstep rule keeps `--examples` / `MANUAL` / `sercon.d.ts` in sync, and the doc map is now the seventh artifact in that chain).

15. Limitations and gotchas

- **No native ES modules at runtime.** All `import` is rewritten to `require` by esbuild. Side-effect-only imports run the module body exactly once per Run.
- **No top-level await inside required modules.** Wrap with `(async () => ... )()` inside the module if you need it.
- **No tsconfig.json honoured.** Module resolution and type checking are independent from any project-level TS config.
- **Per-Run runtime.** Side effects on `globalThis` do not persist between calls to `Run` on the same `Engine`. The `require` *compile* cache is shared; module *exports* are not.
- **Multi-line ESM imports may stress the entry rewriter.** The current rewriter is line-based and handles the common forms; especially gnarly inputs (comments inside the spec list, very unusual whitespace) fall back to executing as-is, which may throw.
- **RegisterConstructor is d.ts-only today.** At runtime it behaves like `Register`. True `new`-able constructor semantics are on the roadmap.
- **HTTP bindings are real network calls.** `net.http.*` uses `net/http` with a 5s timeout. They are not mockable from JS.

See [OUT-OF-SCOPE.md](#) for the active backlog of deferred ideas.

This manual covers sercon v0.13.1. Whenever you add, remove, or change a flag, a binding, or the script API, update this file alongside the help screen (`--help`), the examples walkthrough (`--examples`), and the `CHANGELOG.md`.